Secure Computing Adaptive SoC for Physical AI, Data Center, and Mission Critical Systems

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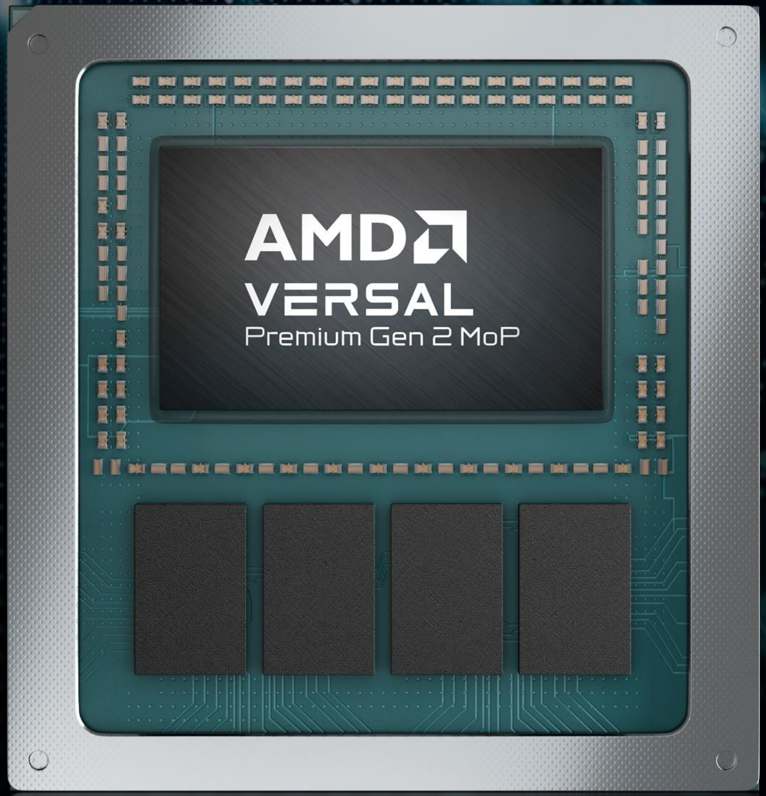


Agenda

Overview	Co-designing Trust, Memory Bandwidth and High-Speed IO for deployable Physical AI platforms
Solution – AMD Versal Premium Gen 2 Family Features	▪ Offload Acceleration Blocks for Host, Network, and Embedded Processing ▪ Adaptive Extensions ▪ Pervasive Security and Confidential Computing
Use Cases	▪ Programmable Adaptive SoC + x86 CPU (Ryzen AI CPU) ▪ CXL Low-Latency Compute – Coherent Acceleration – Memory Expansion ▪ Advanced Medical & Industrial & Avionics Imaging
Summary	
Q&A	

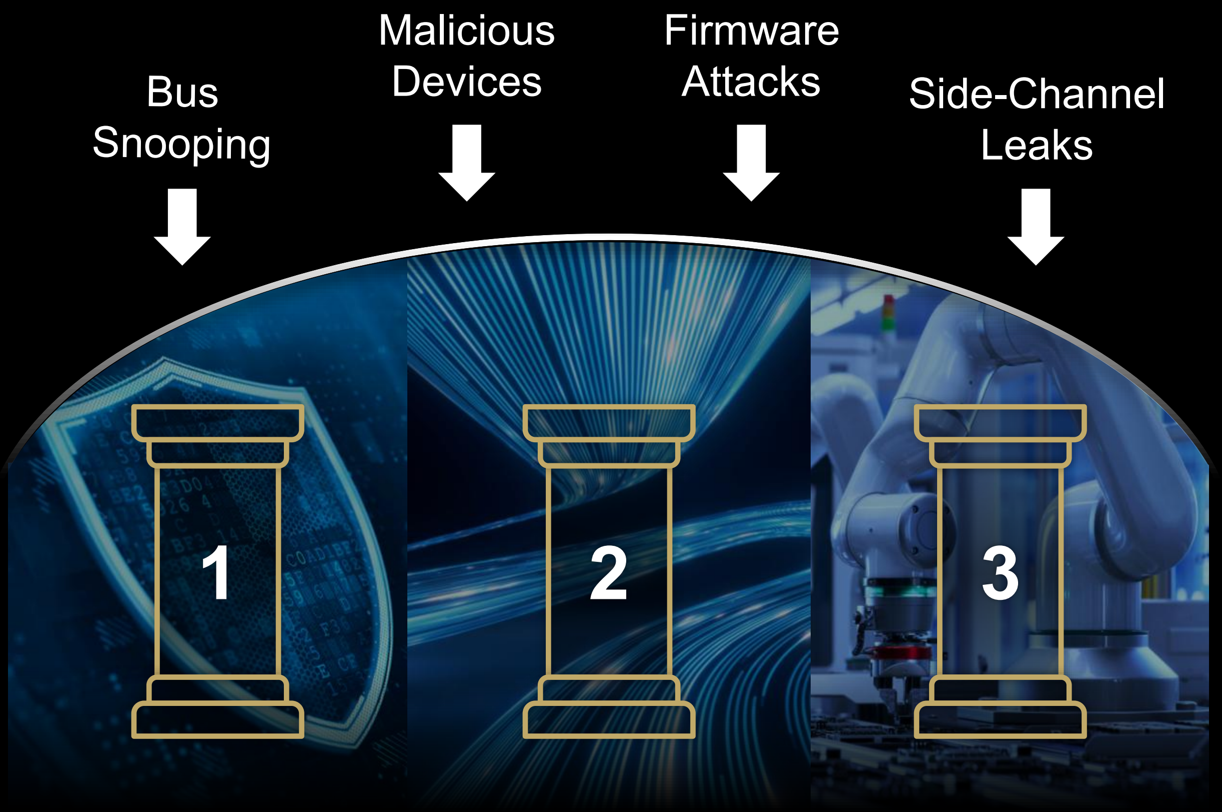
Overview

Motivations: Why the 3 Pillars Must Converge

Three Pillars Must Converge

1. Secure Boot and Crypto Acceleration
2. Secure High-Speed Interconnect & Memory
3. Physical AI for Demanding Environments with Low Latency

Co-designed for Trust, Throughput, and Resilience



Platform Overview

FPGA-Based Adaptive SoC

Compute
Programmable Logic (PL) + Real Time App Cores

Security Primitives

Hardened Interfaces

System Objectives

System Objectives

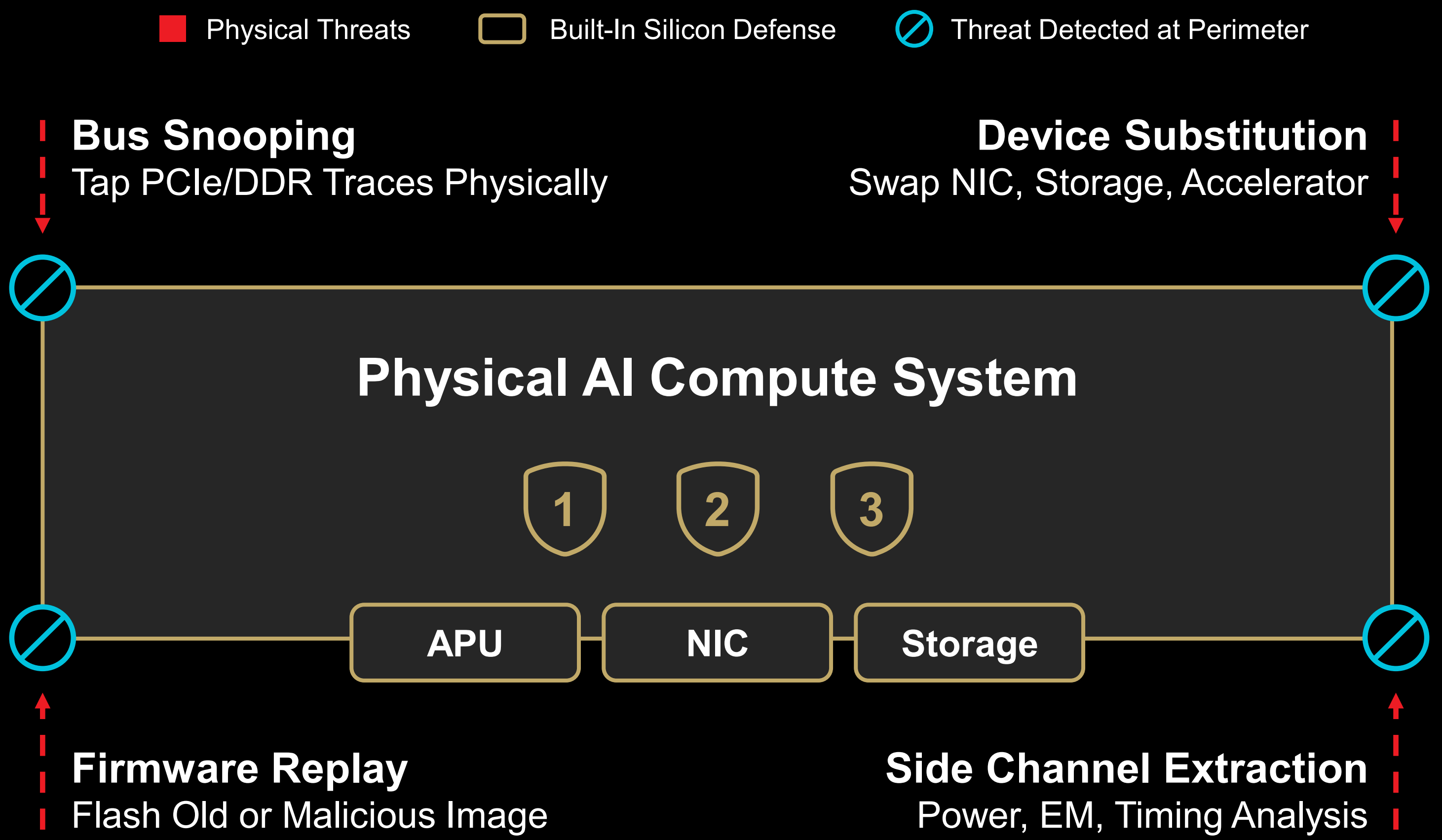
Maximum Throughput for Deploy-ability

No Security Gaps between Subsystems

Isolation Enforced Across Layers

Threat Model for Physical AI System

1. **I/O Fabric**
   PCIe Gen6/7 IDE, DMA Isolation, MACsec
2. **Memory Subsystem**
   Inline Encryption, ECC,TEE
3. **Secure Boot Chain**
   ROM Root of Trust, Signed Stage



Memory Protection Tailored to Different Threat Models

AES-XTS

Multi-Tenant Confidentiality
+
Encrypted data in use

AES-GCM

Protection from Replay attacks

Key Rolling

Execute Long-lived Authenticated Code

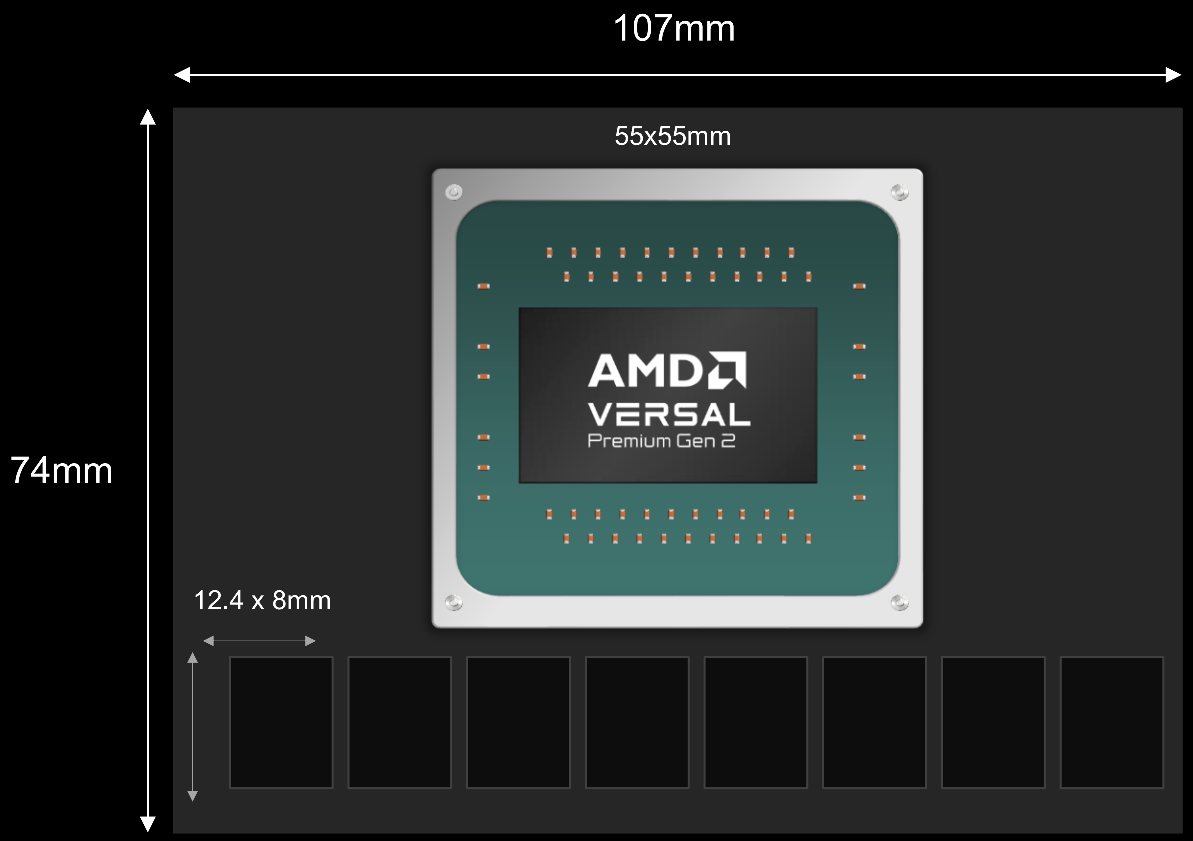
(Prevents Man-in-the-Middle malware attacks)

DPA Counter measures

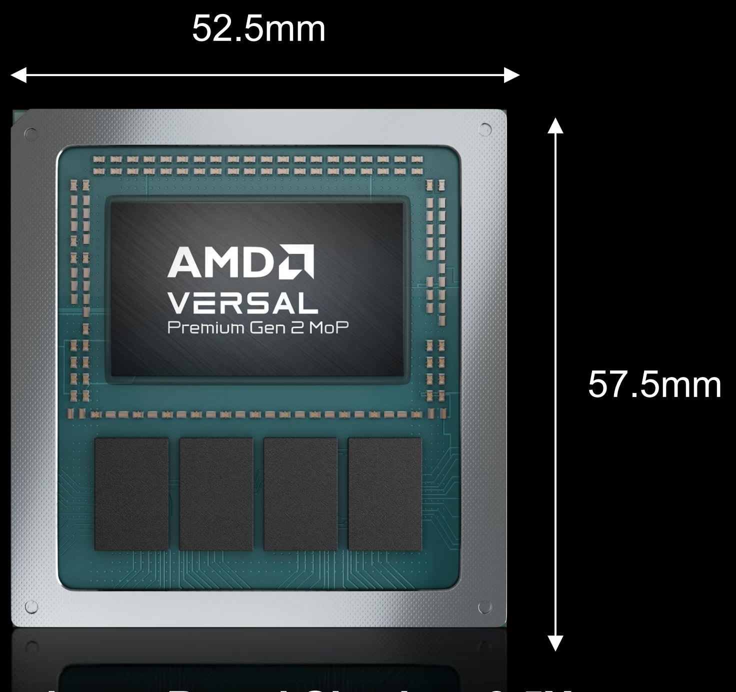
Protection at Remote Physical AI locations

A rich set of Memory Protection Features that can be deployed selectively or in combination across threat models

Why Memory on Package?
Better Density, Higher Reliability and Security



~60%
Less Area

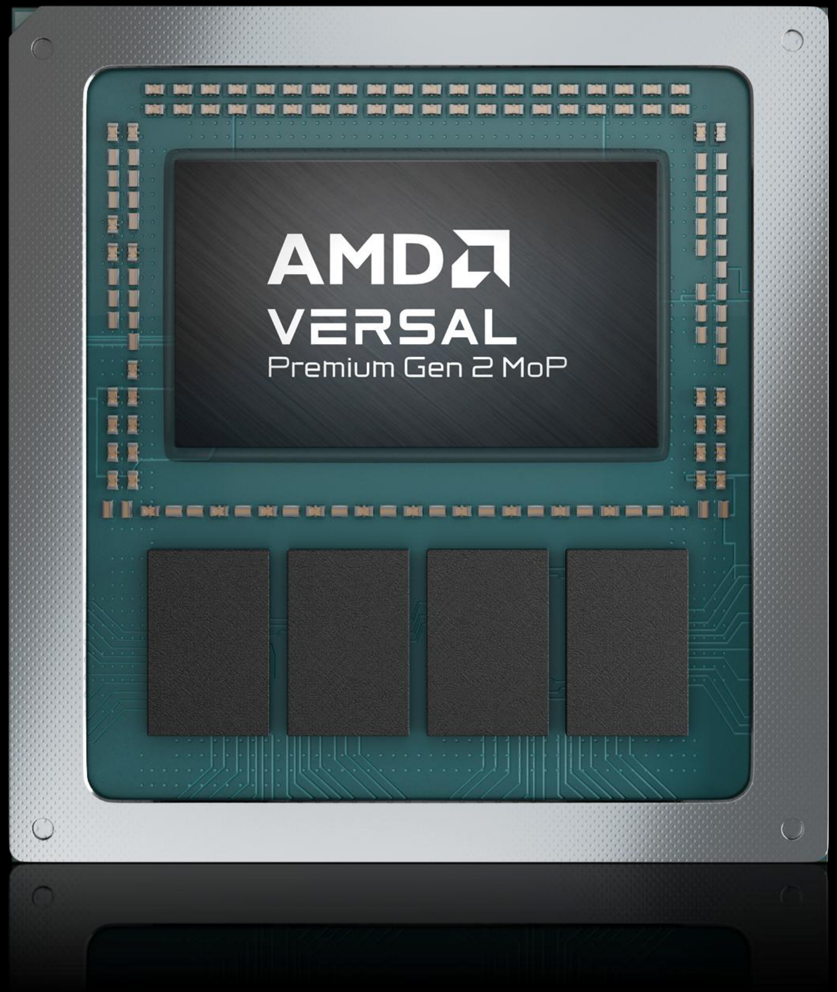


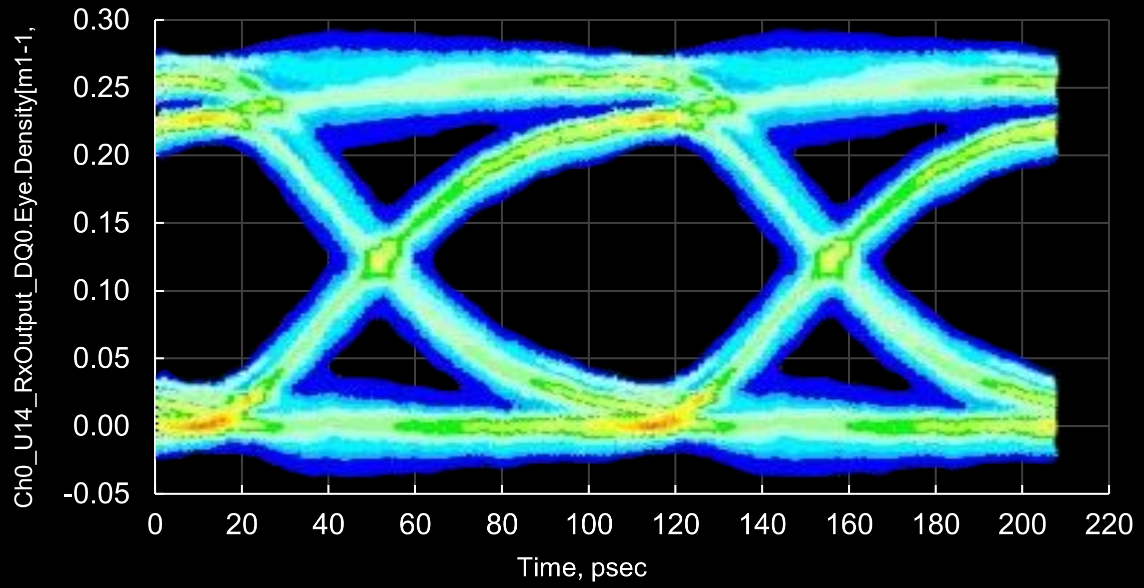
Reduces Board Size by ~2.5X
Fits into PXI, 3U VPX, and PCIe® Form Factors

End Note: VER-061

Secure Memory on Package

Additional Security Feature





- Smaller attack surface, protected contents
- Lower latency, power, and footprint for SWaP-constrained designs
- Always-on encryption protects data-in-use
- -40C to 100C for industrial, airborne, and outdoor use

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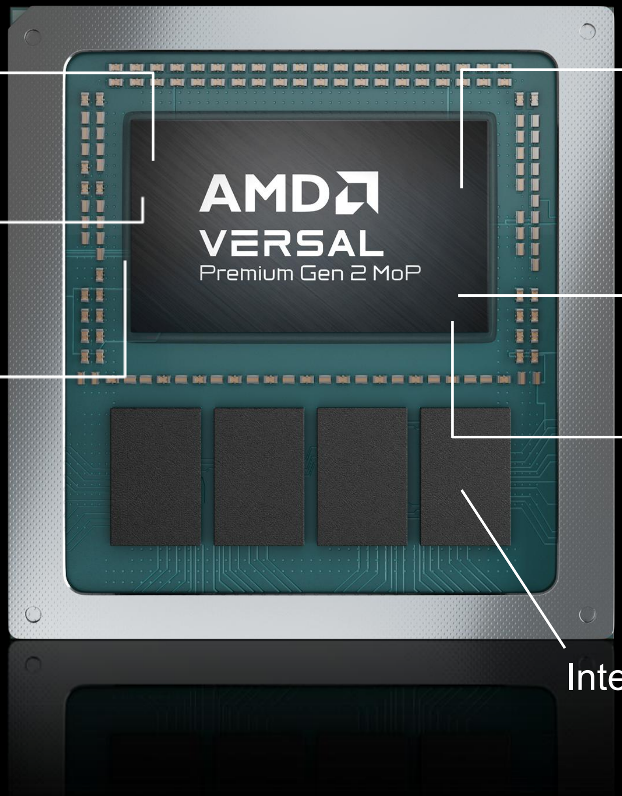
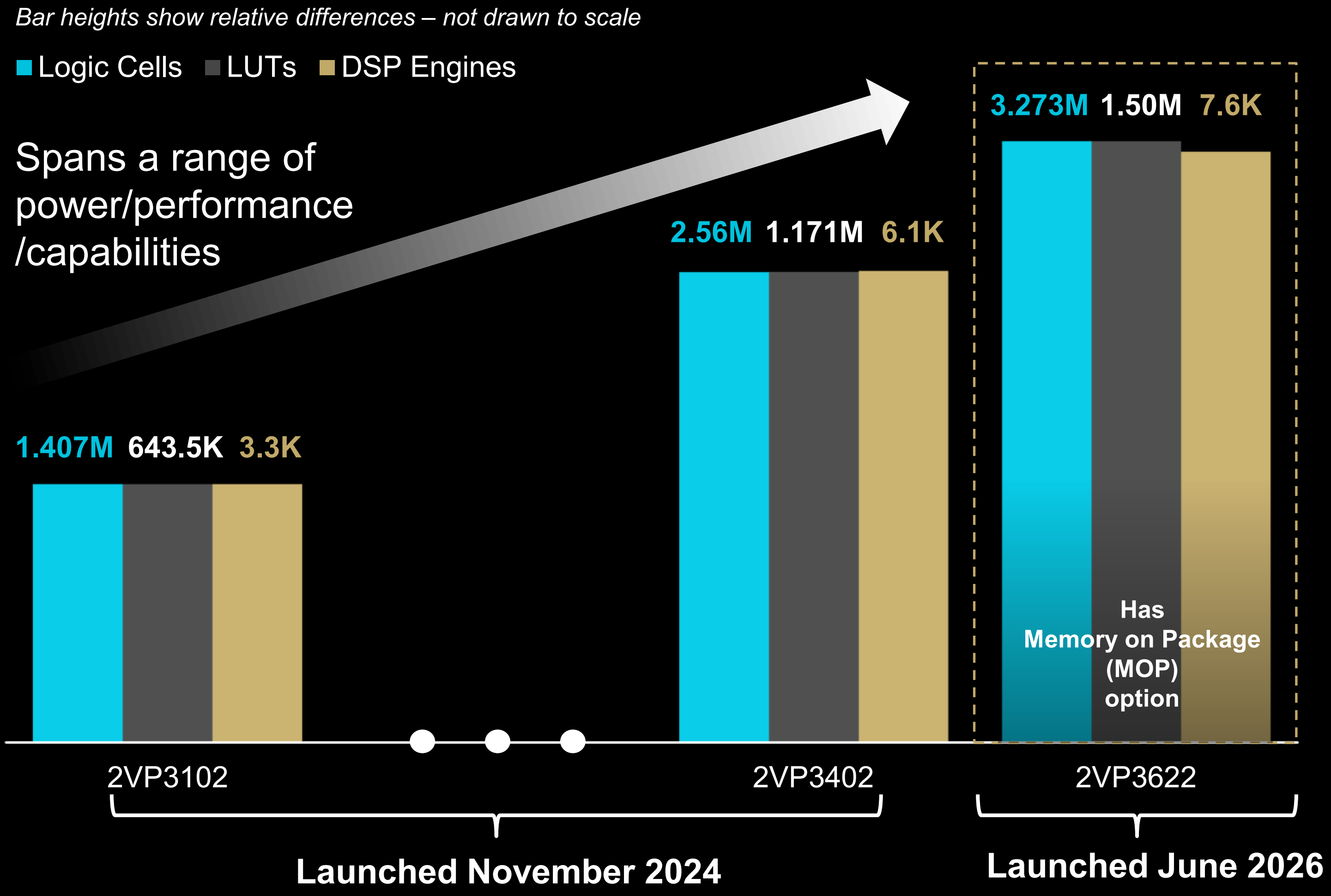
Key Innovations

AMD Versal™ Premium Series Gen 2 Memory on Package



Target Real Time Processing and Data Aggregation on a Physical AI Node with Sensors & Actuators

Versal™ Premium Series Gen 2 Range of Adaptive SoC Products



Logic Cells & LUTs

Benefit Real Time Processing

- Fine-Grained Parallelism
- Increase Pipeline Depth & Throughput
- Improve Resource Mapping
- Reduce Memory Access
- Improve Memory Bandwidth

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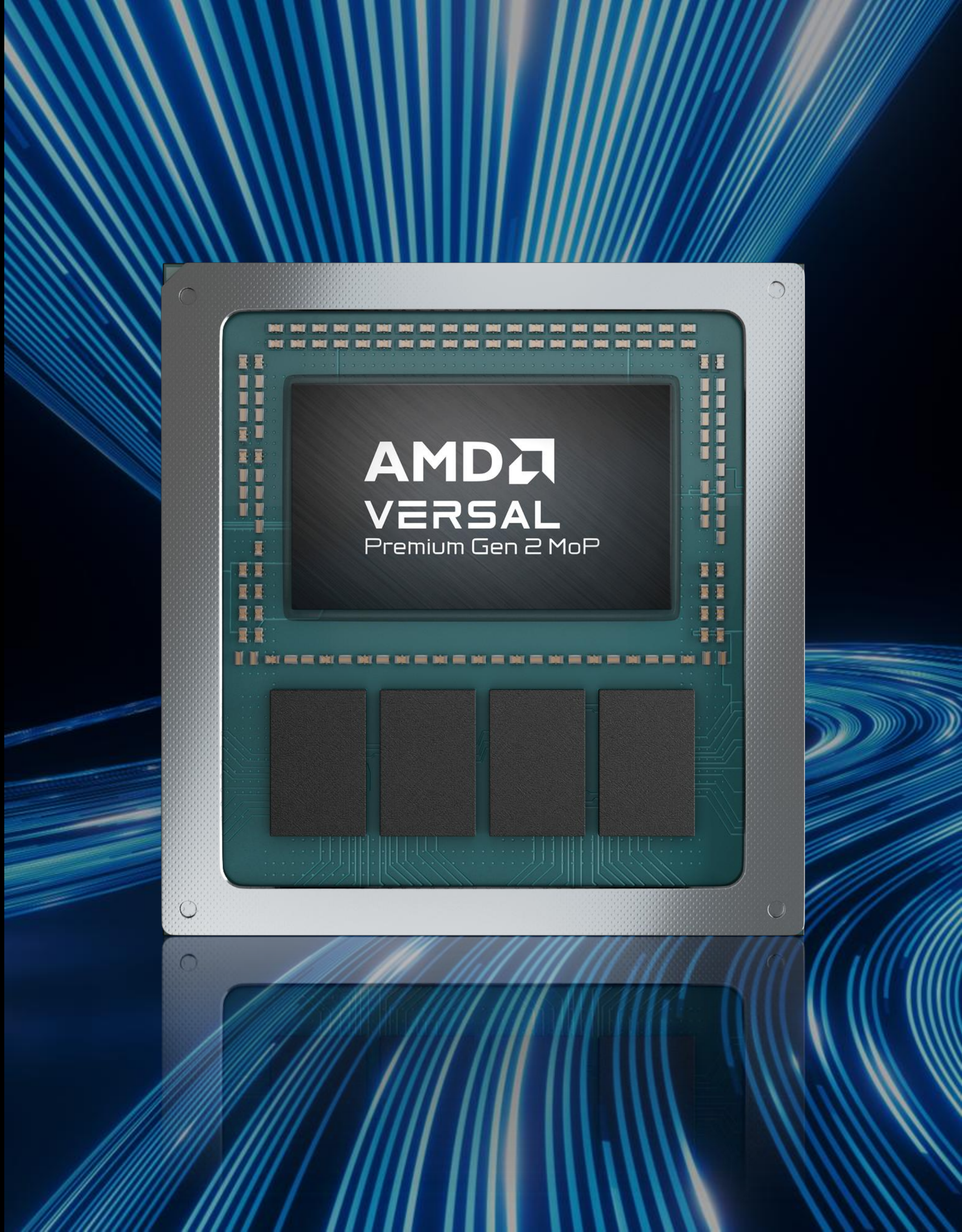
Solutions



Post Quantum Crypto – Not One Size Fits All

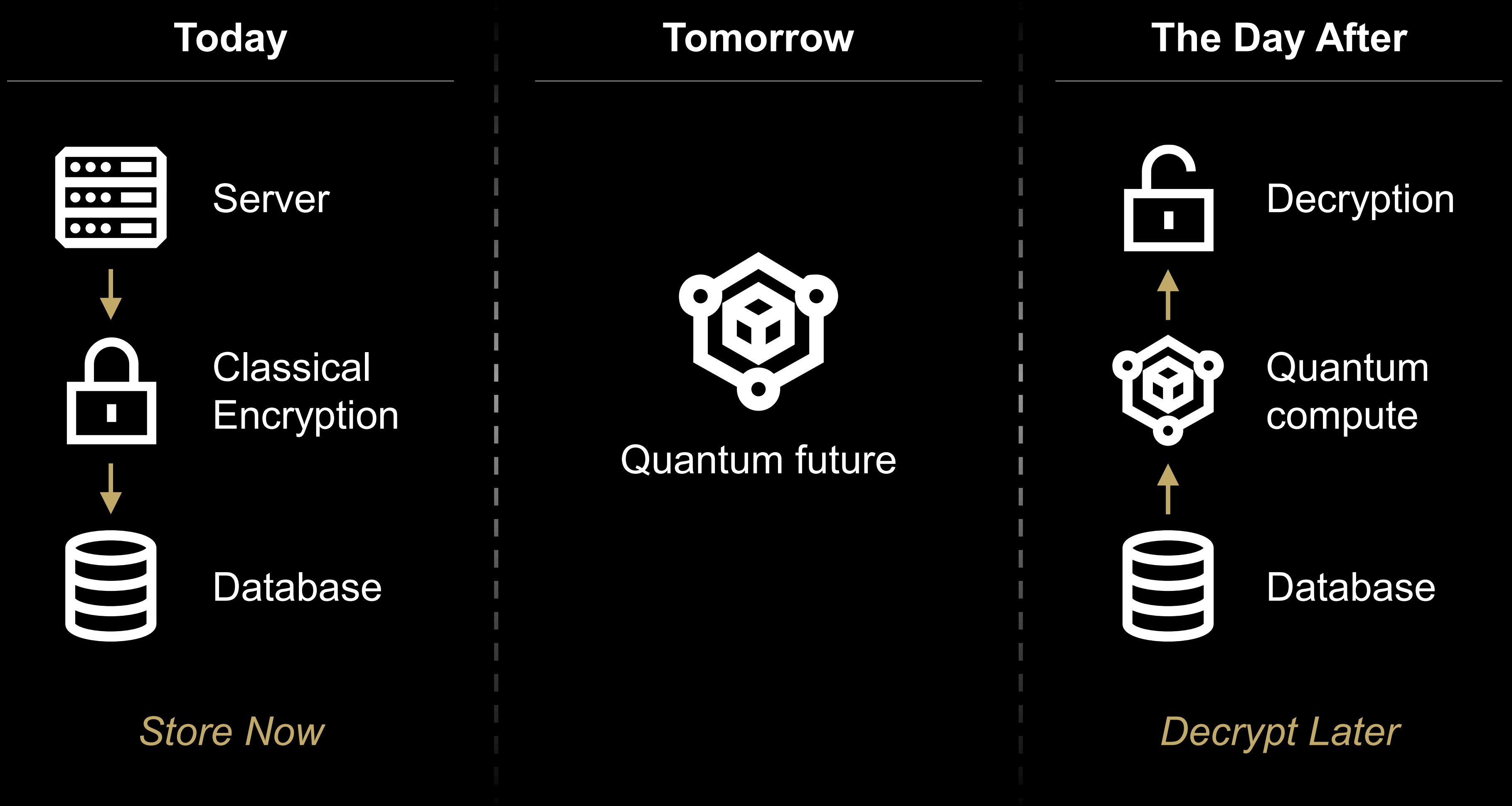
Harvest Now and Decrypt Later

Stay with CNSA 2.0	*Explore new PQC ciphers*	*Gradually transition from classical crypto*
ML-KEM ML-DSA LMS/XMSS	HQC-KEM NTRU (Lattice-based)	Hybrid PQC + ECC e.g., SLH-DSA + ECDSA

Crypto Agility Layer

Global customers require PQC compliance across all regions.

Preparing for Quantum Future



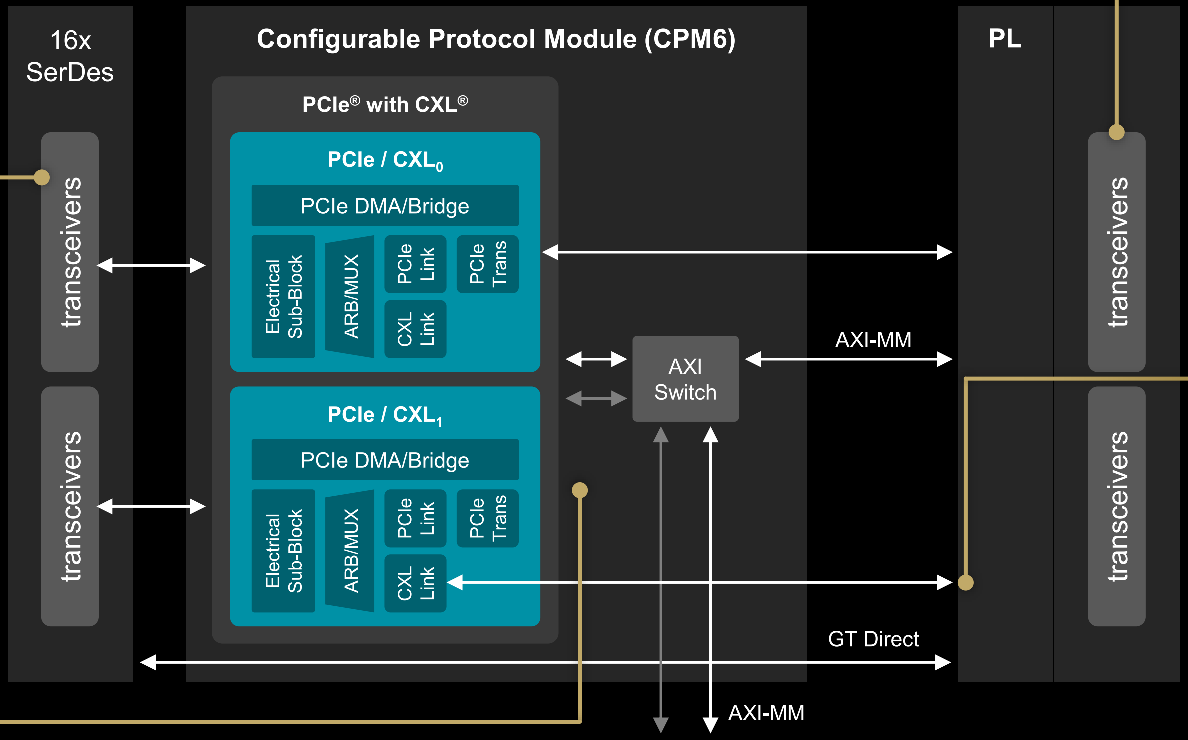
PCIe and CXL

GTM2 Transceiver

- PCIe® Gen 1 to Gen 6
- Up to 64 GT/s
- 2 Quad, 8 Lanes Each

CPM6

- CXL 3.1 ArbMux
- CXL 3.1 Datalink Layer
- CXL 2.1 wide datapath to PL



GTM2 Transceiver

- PCIe® Gen 7 Direct-to-PL
- CXL 4.0 Direct-to-PL
- Up to 128 GT/s

CXL Transaction IP

- CXL.cache & CXL.mem
- Configurable & Expander

PCIe for Gen6 & Gen7 acceleration bandwidth
CXL 3.1 & 4.0 for low latency memory expansion and coherency semantics

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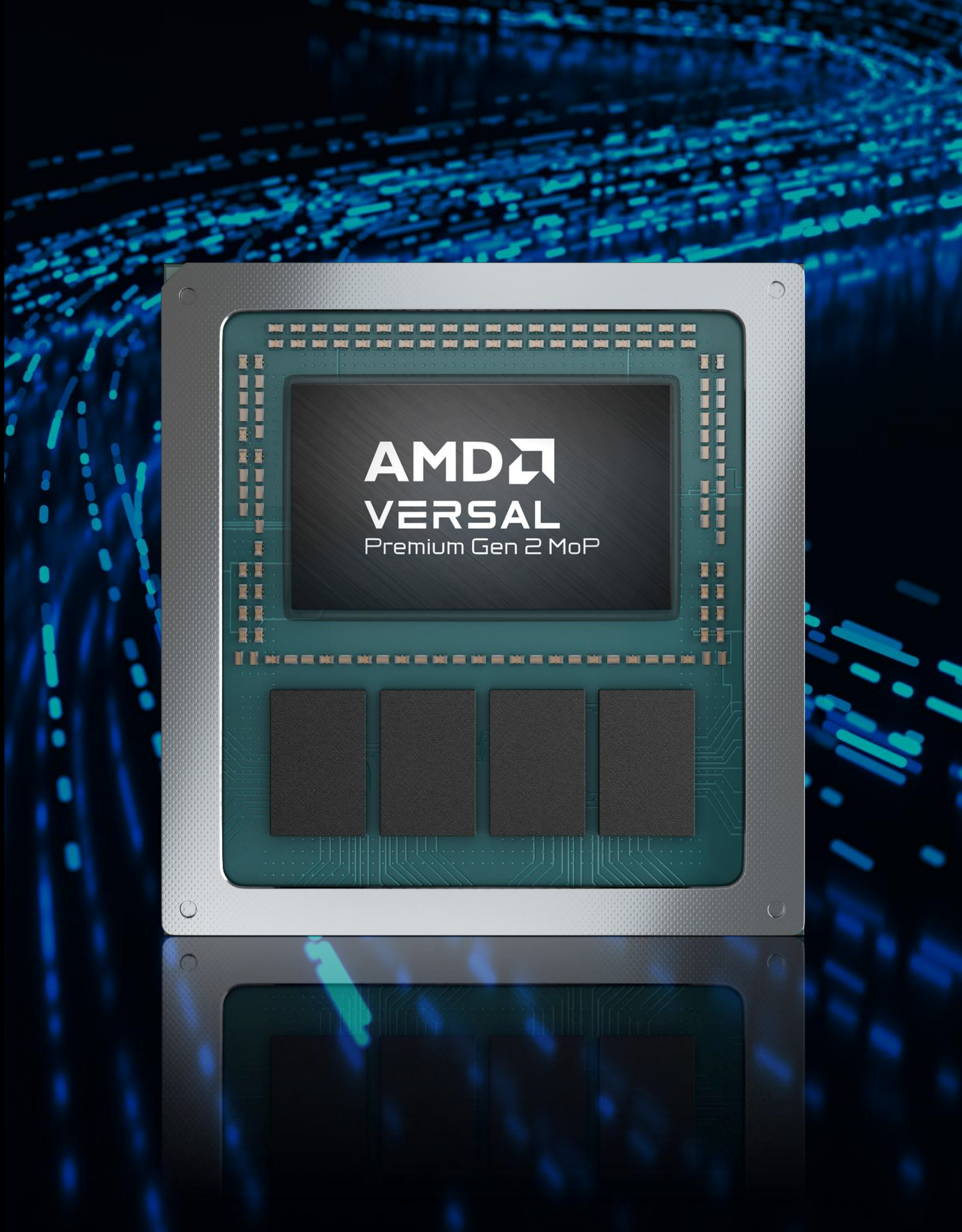
Use Cases



AMD EPYC™ Processor with Versal™ Premium Series Gen 2

System Performance

- PCIe® Gen6 and Gen7 for host-to-accelerator bandwidth
- Offload compute, networking, storage workloads

CXL® for Coherent Memory

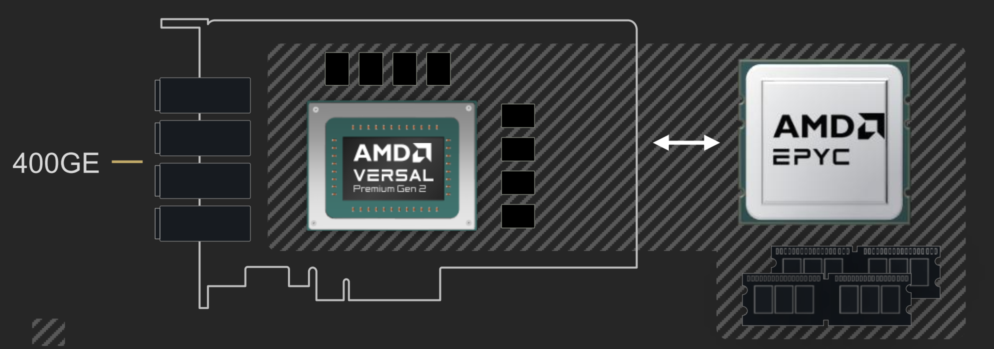
- CXL compliance for both host and accelerator
- Expands available system memory
- Reduces stranded memory waste

End-to-End Security

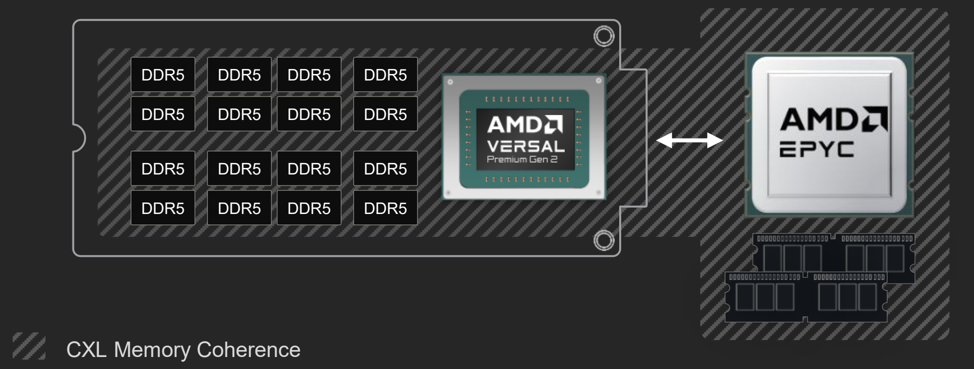
- PCIe IDE for host-to-accelerator security
- Encrypted memory in flight

Tightly coupled pairing with EPYC to create a low-latency, power-efficient solution

Host-to-Accelerator



Host-to-Memory Expansion Module



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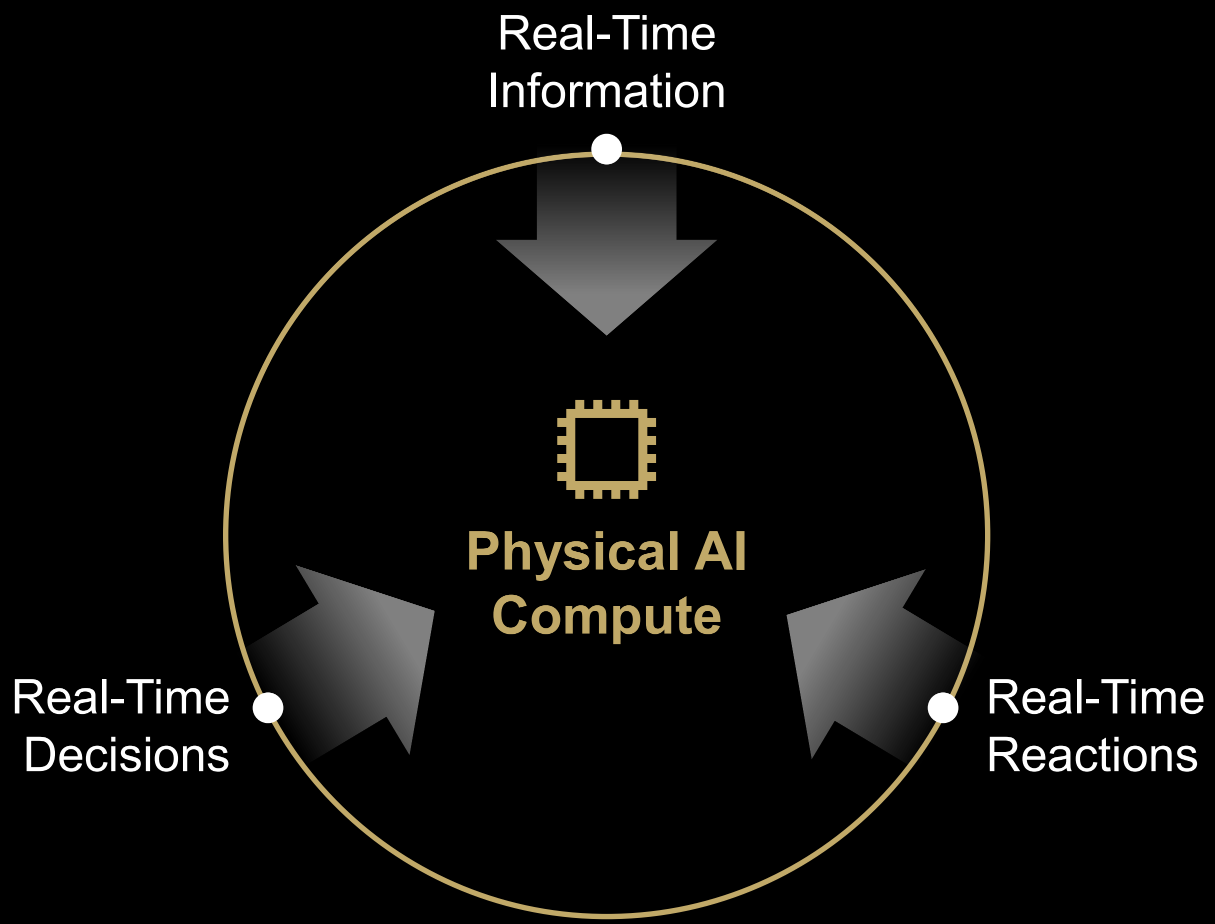
Characteristics of a Physical AI Compute Platform



All work in concert – working all the time

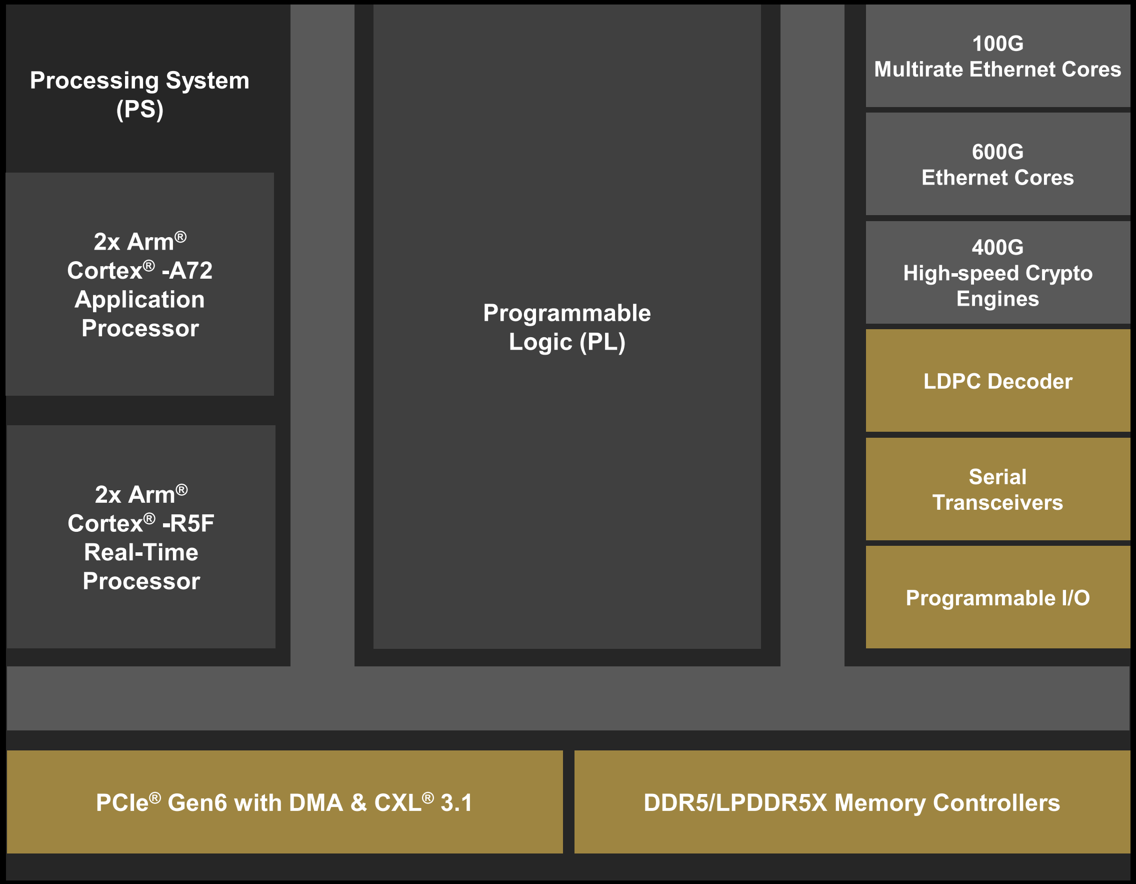
The Physical AI world never sleeps

Feature Values For Physical AI Applications

Physical AI ***Data Bottleneck***

Central Aggregator for Physical AI data Conditioning	→	Programmable Logic flexibility for different data conditioning algorithms
Real-time processor	→	Programmable Logic flexibility for short, real-time decision loops
Secure Data Pipeline for multi-sensor environments	→	Programmable Logic flexibility for changing sensor interfaces and sensor types

Versal Premium Gen 2



Physical AI for Industrial and Robotics

Real Time Input

- Sensors
- Analog Input
- Digital Output / Actuators

Real Time Analysis

Versal Premium Gen2

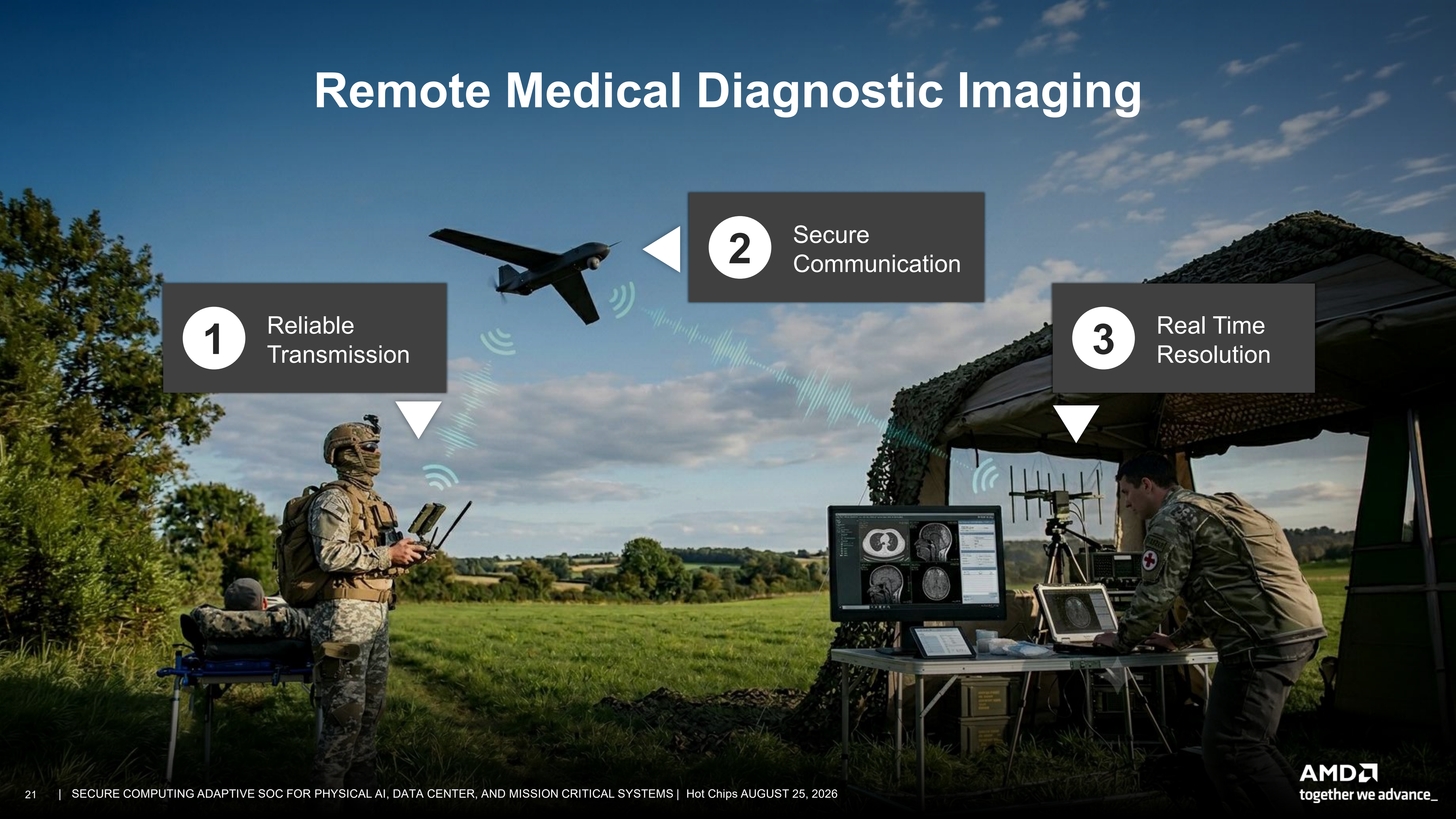
- Real-time I/O
- 10 μs loops
- Protocol IP
- Diagnostics

Real Time Action

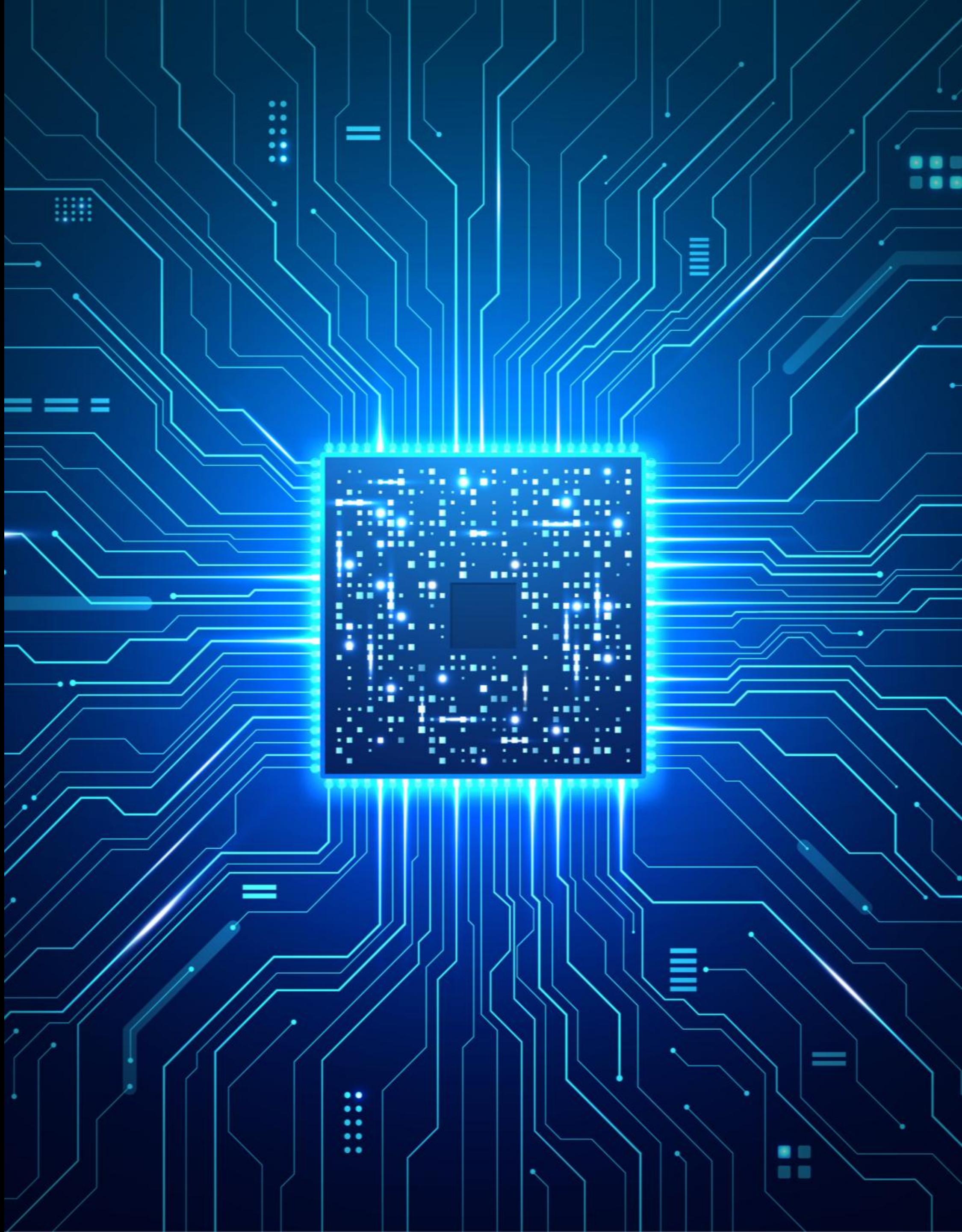
Ryzen AI Embedded

- RTOS/OS
- 10-40 ms
- Apps
- Networking

Remote Medical Diagnostic Imaging



Summary



Summary

Think Ahead to Protect Your Data Now

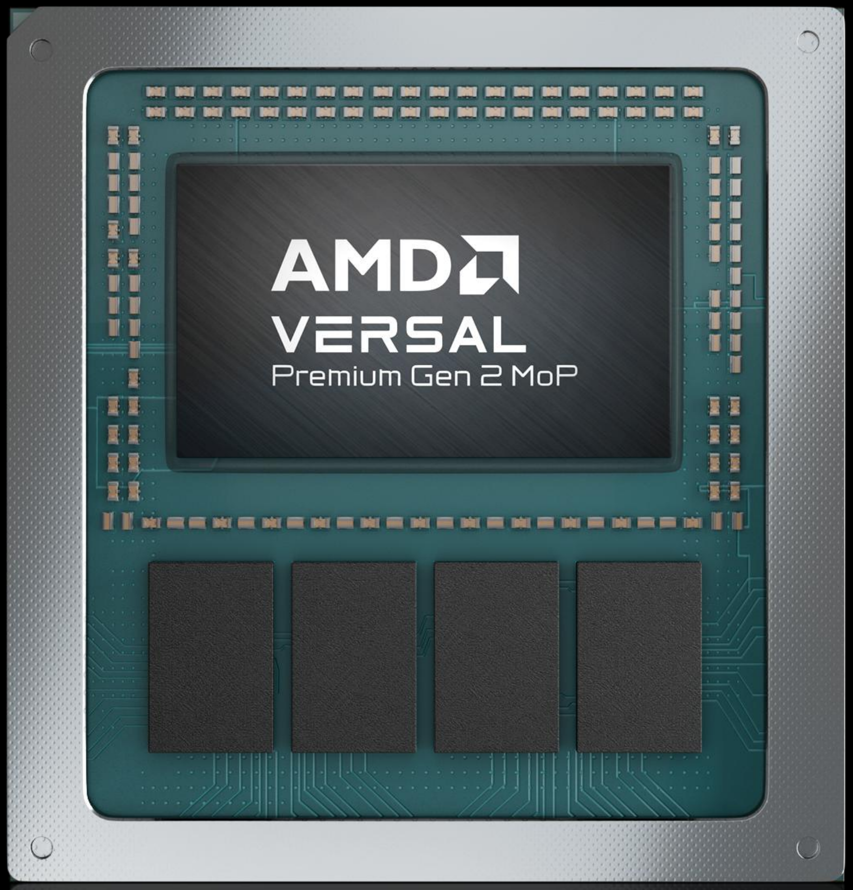
- Not Short Term
- Think Long Term!!

Information Changes in Real Time

- Information must be processed in Real Time
- Decision must be made in Real Time and with Low-Latency
- Action must be taken in Real Time

System Differentiator

- Co-design of trust anchor
- Secure memory and data in motion
- High-speed interconnect for sensors and communication



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End notes

VER-058
Based on AMD internal analysis of Versal Premium Series Gen 2 device DDR/LPDDR memory interface specifications vs. comparable competitive devices, as of July 2024.

VER-059
Based on AMD internal analysis of the total memory bandwidth (CXP3.1 and LPDDR5X memory components) available with AMD Versal Premium Series Gen 2 devices vs. the same devices with LPDDR5X memory alone Memory bandwidth will vary based on system configuration and other factors.

VER-061
Based on AMD internal analysis of AMD Versal Premium Series Gen 2 devices vs. previous generation Versal Premium Series, as of July 2024. Actual DSP-to-logic Ratio will vary based on system configuration and other factors.
Compared to total area consumed using 2VP3602-VSVC3340 and eight LPDDR5 devices (107mm x 74mm estimated)
Eliminates board level DRAM interface; 0.92mm pitch vs. 0.8mm pitch on LPDDR6x devices; reduces board layers and design & verification effort.

VER-062
Image generated on slide 21 using Google Gemini.

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